西南交通大學

有限元法及应用

Finite Element Method and Application

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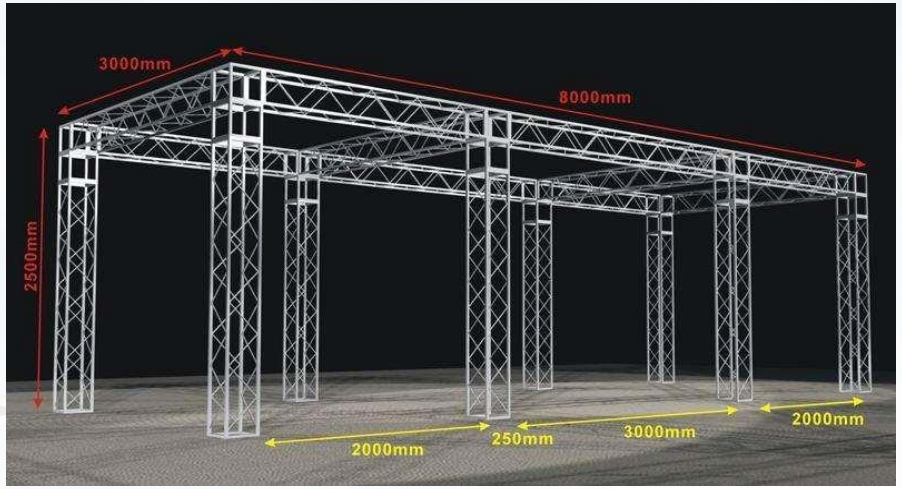
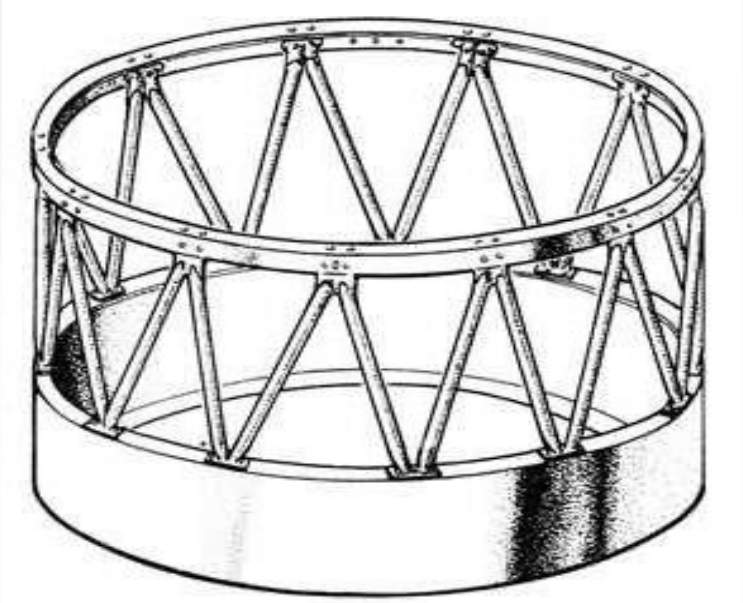
第2章 杆系结构的有限元法

工程背景

钢架、桁架的应用。





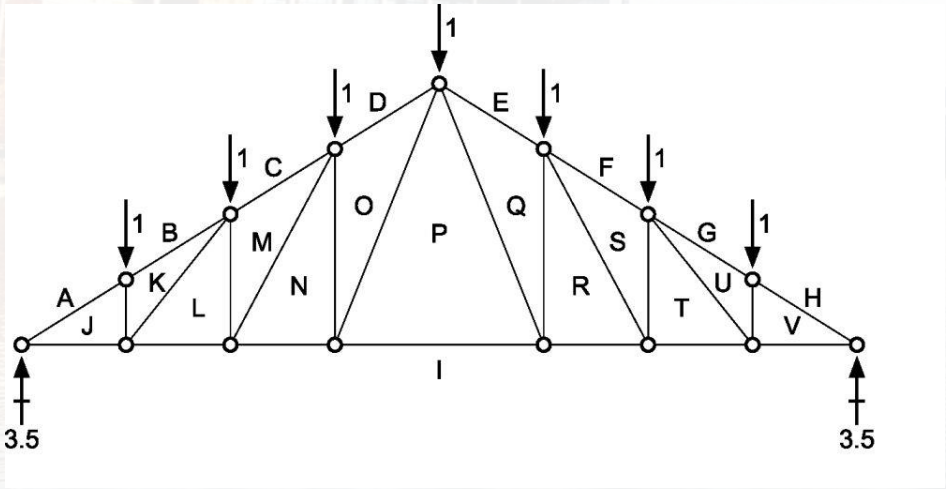
2.1 基本概念

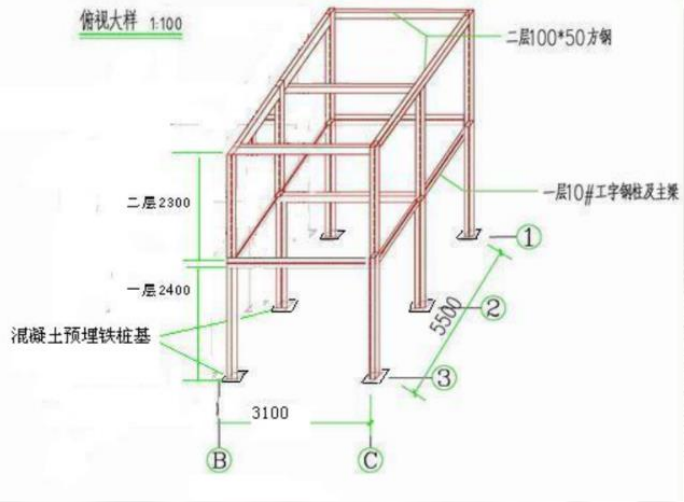
节点和单元

杆系结构的单元类型分为杆单元和梁单元，二者均为线单元。

杆单元的节点仅传递力而不传递力矩，其节点位移只有沿坐标系各个轴向的线位移；

梁单元的节点不仅传递力，而且传递力矩，其节点位移有沿坐标系各个轴向的线位移和绕坐标系各轴旋转的角位移。

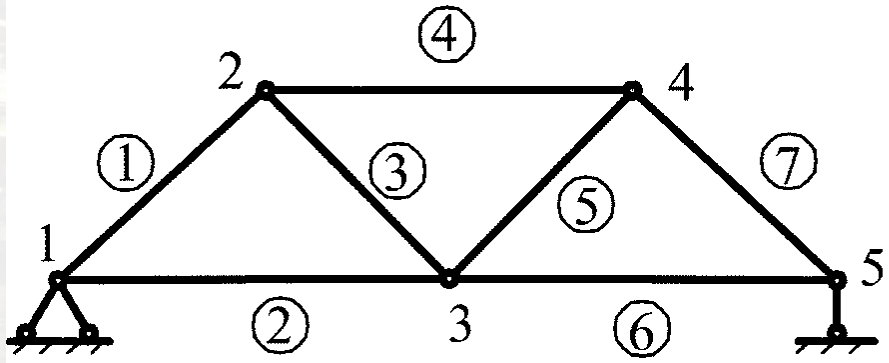




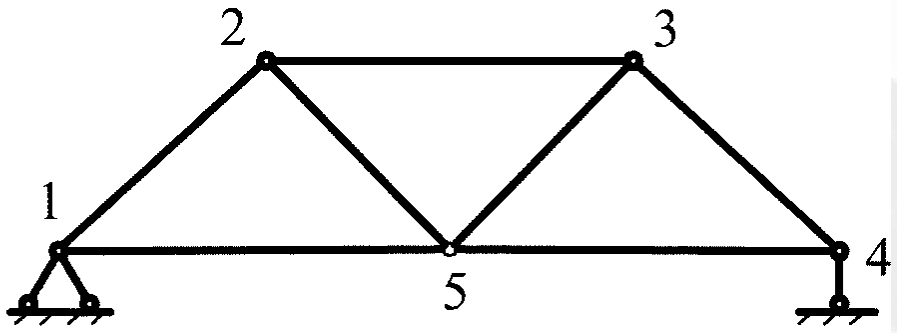
节点编号原则：编号不能重复且不能遗漏，相邻节点编号差尽量小。

单元编号原则：编号不能重复且不能遗漏，相邻节点编号差尽量小。

商业有限元软件可自动完成。



(a)



(b)

节点和单元编号方法

2.1 基本概念

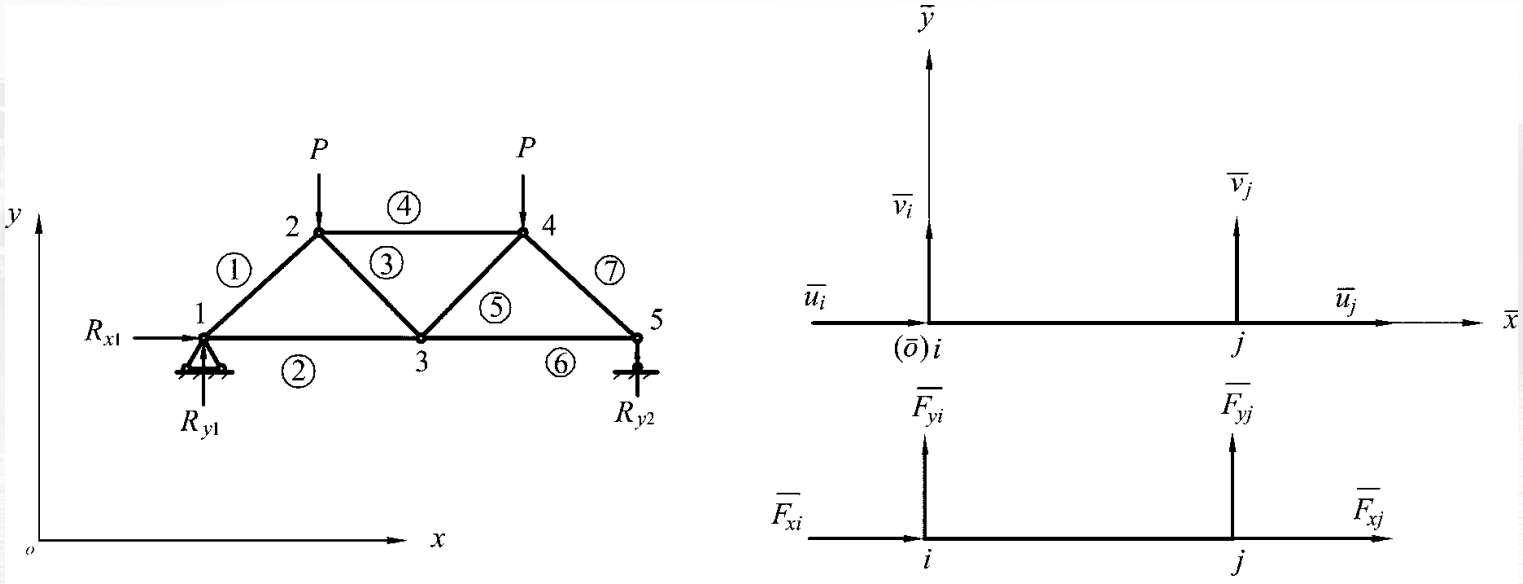
边界条件是计算模型部分节点上的约束和荷载的总称。

整体坐标系是指标识计算模型空间位置关系的坐标系。

节点坐标是表示节点在整体坐标系下的空间位置。

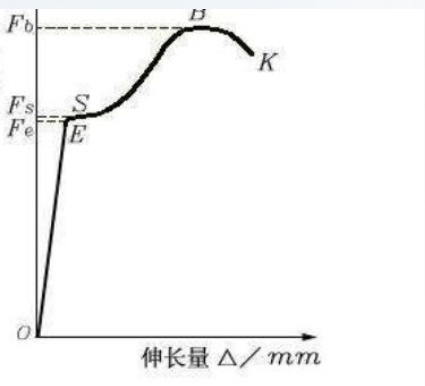
单元节点信息是指每个单元所包含的节点数量。

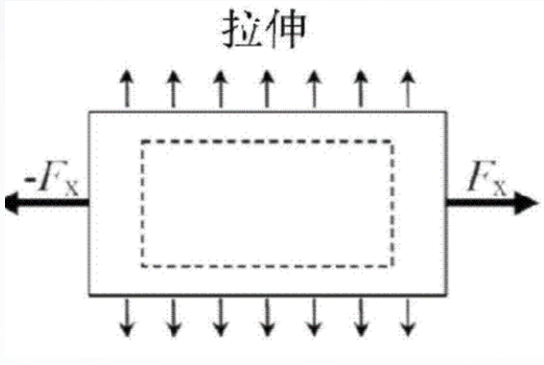
单元局部坐标系



2.1 基本概念

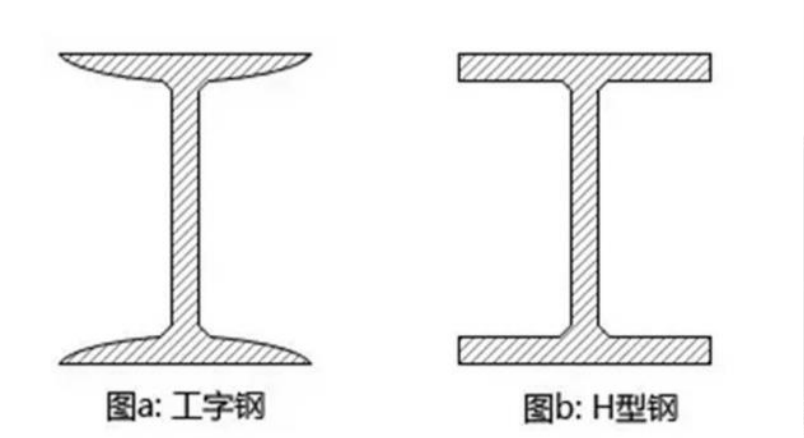
单元材料特性是指每个单元的材料特性参数。对于线性弹性结构有限元分析，单元的材料特性参数主要包括弹性模量、泊松比和密度。





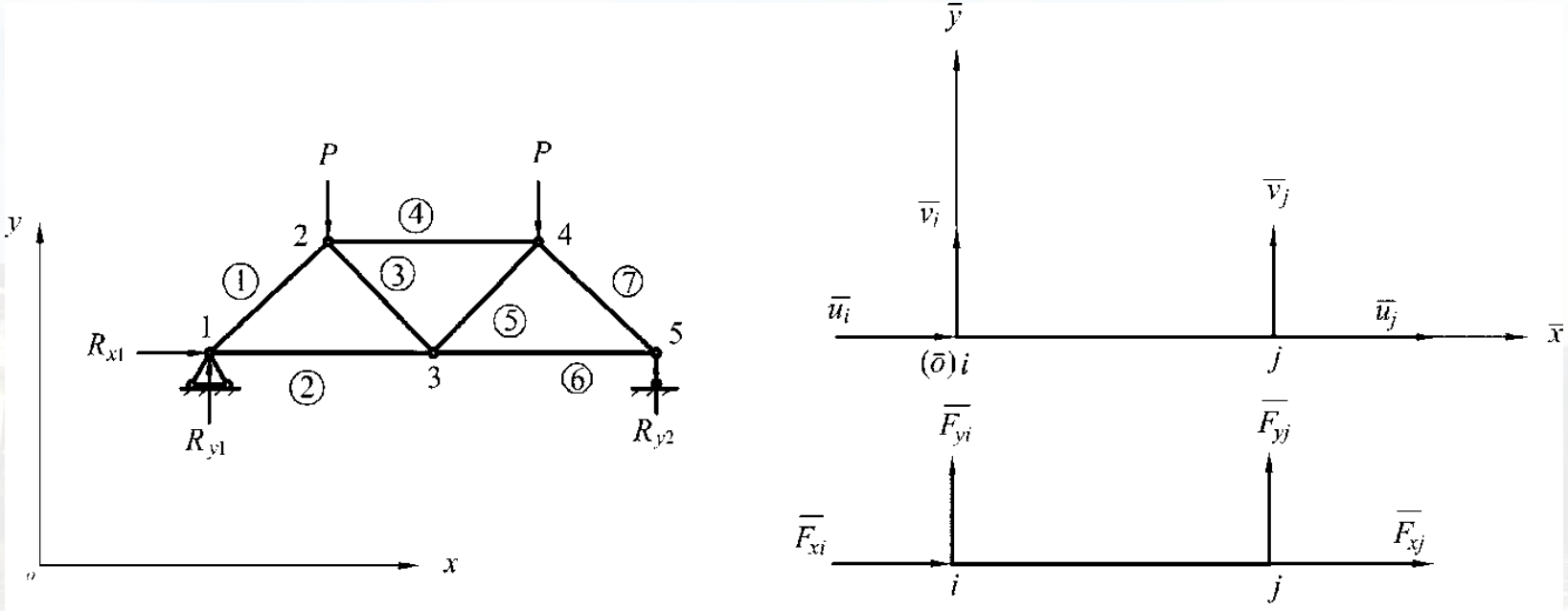
杆单元的截面参数包含截面面积和截面形心位置。

梁单元的截面形状参数包含横截面面积、截面心形位置、剪切中心位置、截面剪切面积（考虑剪切对变形的影响时输入）、抗弯惯性矩和极惯性矩等。



2.2 平面桁架

- 确定单元类型：平面杆单元
- 单元编号、节点编号



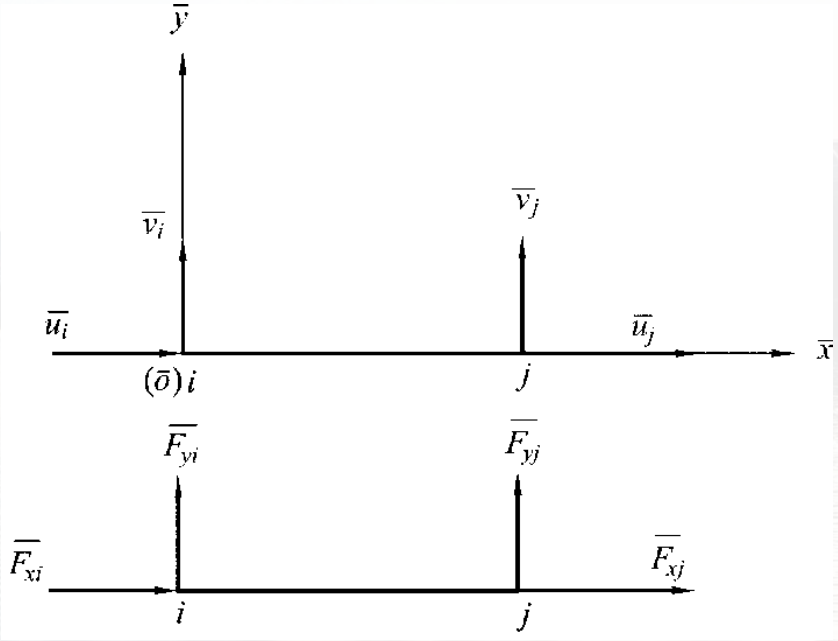
2-2 平面桁架有限元模型

2.2.1 局部坐标系下的单元刚度矩阵

2.2.1.1 单元局部坐标系下的节点位移和节点力

$$\overline{\delta_i}=\begin{bmatrix}\overline{u_i}\\ \overline{v_i}\end{bmatrix}\qquad \overline{\delta_j}=\begin{bmatrix}\overline{u_j}\\ \overline{v_j}\end{bmatrix}\qquad \overline{F_i^e}=\begin{bmatrix}\overline{F_{xi}}\\ \overline{F_{yi}}\end{bmatrix}\qquad \overline{F_j^e}=\begin{bmatrix}\overline{F_{xj}}\\ \overline{F_{yj}}\end{bmatrix}$$

$$\overline{\delta^e}=\begin{bmatrix}\overline{\delta_i}\\ \overline{\delta_j}\end{bmatrix}=\begin{bmatrix}\overline{u_i}\\ \overline{v_i}\\ \overline{u_j}\\ \overline{v_j}\end{bmatrix}$$



2.2.1.1 单元局部坐标系下的节点位移和节点力

$$\overline{\delta^e}=[\overline{\delta_i}\quad \overline{\delta_j}]^T=[\overline{u_i}\quad \overline{v_i}\quad \overline{u_j}\quad \overline{v_j}]^T \qquad (2-1)$$

$$\overline{F^e}=[\overline{F_i^e}\quad \overline{F_j^e}]^T=[\overline{F_{xi}}\quad \overline{F_{yi}}\quad \overline{F_{xj}}\quad \overline{F_{yj}}]^T \qquad (2-2)$$

2.2.1.2 单元节点位移与节点力的关系

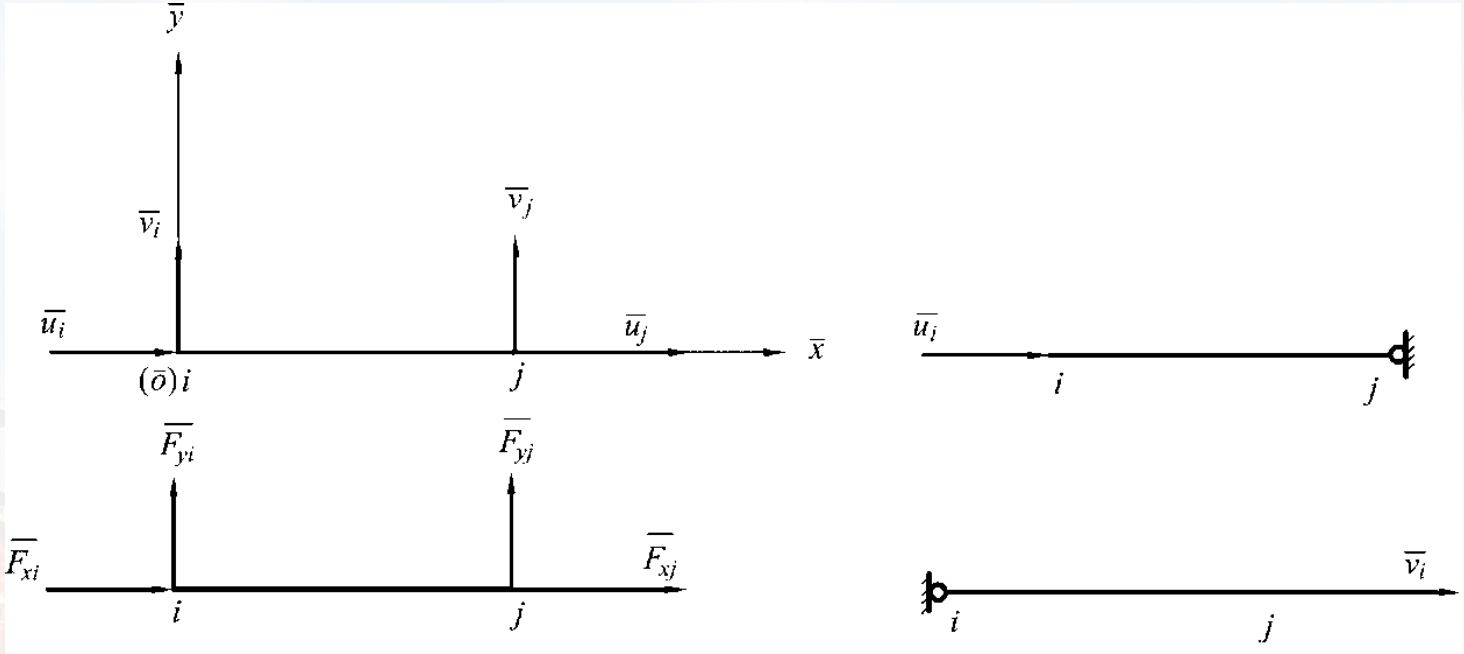


图2—3 平面杆单元受力分析

回顾《材料力学》单向拉伸:

$$\Delta l = \frac{Fl}{EA}$$

杆件处于静态平衡状态，必有

$$\overline{F}_{xj} = -\overline{F}_{xi}$$

因此，轴向力与轴向位移的关系为

$$\overline{F}_{xi} = \frac{EA}{l}\left(\overline{u}_i - \overline{u}_j\right) = \frac{EA}{l}\overline{u}_i - \frac{EA}{l}\overline{u}_j \qquad (1)$$

$$\overline{F}_{xj} = -\overline{F}_{xi} = -\frac{EA}{l}\overline{u}_i + \frac{EA}{l}\overline{u}_j \qquad (2)$$

二力杆，垂向载荷恒等于0，因此必有

$$\overline{F}_{yi} = 0\overline{u}_i + 0\overline{v}_i + 0\overline{u}_j + 0\overline{v}_j \qquad (3)$$

$$\overline{F}_{yj} = 0\overline{u}_i + 0\overline{v}_i + 0\overline{u}_j + 0\overline{v}_j \qquad (4)$$

写成矩阵形式:

$$\begin{bmatrix} \overline{F}_{xi} \\ \overline{F}_{yi} \\ \overline{F}_{xj} \\ \overline{F}_{yj} \end{bmatrix} = \begin{bmatrix} \frac{EA}{l} & 0 & -\frac{EA}{l} & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{EA}{l} & 0 & \frac{EA}{l} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \overline{u}_i \\ \overline{v}_i \\ \overline{u}_j \\ \overline{v}_j \end{bmatrix} \qquad (2-3)$$

即:

$$\overline{F}^e = \overline{K}^e \overline{\delta}^e \qquad (2-4)$$

其中，

$$\overline{K^e}=\begin{bmatrix}\overline{k_{11}} & \overline{k_{12}} & \overline{k_{13}} & \overline{k_{14}}\\ \overline{k_{21}} & \overline{k_{22}} & \overline{k_{23}} & \overline{k_{24}}\\ \overline{k_{31}} & \overline{k_{32}} & \overline{k_{33}} & \overline{k_{34}}\\ \overline{k_{41}} & \overline{k_{42}} & \overline{k_{43}} & \overline{k_{44}}\end{bmatrix}=\begin{bmatrix}\frac{EA}{l} & 0 & -\frac{EA}{l} & 0\\ 0 & 0 & 0 & 0\\ -\frac{EA}{l} & 0 & \frac{EA}{l} & 0\\ 0 & 0 & 0 & 0\end{bmatrix} \qquad (2-5)$$

$$=\frac{EA}{l}\begin{bmatrix}1 & 0 & -1 & 0\\ 0 & 0 & 0 & 0\\ -1 & 0 & 1 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$$

2.2.1.3 单元刚度矩阵的性质

（1）单元刚度矩阵是单元上由节点位移向量求节点力向量的转换矩阵。

$$\overline{F^e} = \overline{K^e}\,\overline{\delta^e}$$

（2）刚度矩阵中各元素的力学意义为：矩阵中的元素均是单位节点位移所引起的节点力的数值，所以称它们为刚度系数。刚度矩阵中每行或每列元素的力学意义是：同一行的4个元素是4个节点位移对同一个节点力的影响系数(或称贡献系数)；同一列的4个元素是同一个节点位移对4个节点力的影响系数。

$$\begin{bmatrix} \overline{F_{xi}} \\ \overline{F_{yi}} \\ \overline{F_{xj}} \\ \overline{F_{yj}} \end{bmatrix} = \begin{bmatrix} \frac{EA}{l} & 0 & -\frac{EA}{l} & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{EA}{l} & 0 & \frac{EA}{l} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \overline{u_i} \\ \overline{v_i} \\ \overline{u_j} \\ \overline{v_j} \end{bmatrix}$$

（3）奇异性：单元刚度矩阵是奇异矩阵，即单元刚度矩阵对应行列式值等于零。若把式（2-5）表示的单元刚度矩阵的第3行加到第1行上去，也能使第1行的元素全部为零，因此，单元刚度矩阵对应的行列式值等于零。单元刚度矩阵的奇异性反映了矩阵中还没有考虑到单元两端与整个结构的联系，所以可以产生任意的刚体位移。

$$\overline{K}^{e}=\frac{EA}{l}\begin{bmatrix}1 & 0 & -1 & 0\\ 0 & 0 & 0 & 0\\ -1 & 0 & 1 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$$

（4）分块性：在式（2-1）中，单元节点位移向量是由两个子向量组成的，每一子向量包含一个节点上的两个位移分量；在式（2-2）中，单元节点力向量也是由两个子向量组成的，每一个子向量包含一个节点上的两个力分量；同样在式（2-3）中，对单元刚度矩阵的行与列按同样的原则划分，这样就得到单元刚度矩阵的分块形式，若将其中每一子块记作, 则式（2-5）可写为

$$\begin{bmatrix} \overline{F_{xi}} \\ \overline{F_{yi}} \\ \overline{F_{xj}} \\ \overline{F_{yj}} \end{bmatrix} = \begin{bmatrix} \frac{EA}{l} & 0 & -\frac{EA}{l} & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{EA}{l} & 0 & \frac{EA}{l} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \overline{u_i} \\ \overline{v_i} \\ \overline{u_j} \\ \overline{v_j} \end{bmatrix}$$

$$\overline{K^e} = \begin{bmatrix} \overline{K_{ii}^e} & \overline{K_{ij}^e} \\ \overline{K_{ji}^e} & \overline{K_{jj}^e} \end{bmatrix} \tag{2-6}$$

$$\begin{bmatrix} \overline{F_i^e} \\ \overline{F_j^e} \end{bmatrix} = \begin{bmatrix} \overline{K_{ii}^e} & \overline{K_{ij}^e} \\ \overline{K_{ji}^e} & \overline{K_{jj}^e} \end{bmatrix} \begin{bmatrix} \overline{\delta_i} \\ \overline{\delta_j} \end{bmatrix} \tag{2-7}$$

$$\left.\begin{aligned} \overline{F_i^e} &= \overline{K_{ii}^e}\,\overline{\delta_i} + \overline{K_{ij}^e}\,\overline{\delta_j} \\ \overline{F_j^e} &= \overline{K_{ji}^e}\,\overline{\delta_i} + \overline{K_{jj}^e}\,\overline{\delta_j} \end{aligned}\right\} \tag{2-8}$$

（5）对称性：根据力学中力的互等定理和单元刚度系数的力学意义，有 $\overline{K_{ij}} = \overline{K_{ji}}$，此可知单元刚度短阵为对称矩阵。

$$\overline{K^e} = \overline{K^e}^T$$

2.2.2 整体坐标系下的单元刚度矩阵

2.2.2.1 坐标变换

$$\delta_i = \begin{bmatrix} u_i \\ v_i \end{bmatrix} \quad \delta_j = \begin{bmatrix} u_j \\ v_j \end{bmatrix} \qquad F_i^e = \begin{bmatrix} F_{xi} \\ F_{yi} \end{bmatrix} \quad F_j^e = \begin{bmatrix} F_{xj} \\ F_{yj} \end{bmatrix}$$

$$\delta^e = [\delta_i \quad \delta_j]^T = [u_i \quad v_i \quad u_j \quad v_j]^T$$

$$F^e = [F_i^e \quad F_j^e]^T = [F_{xi} \quad F_{yi} \quad F_{xj} \quad F_{yj}]^T$$

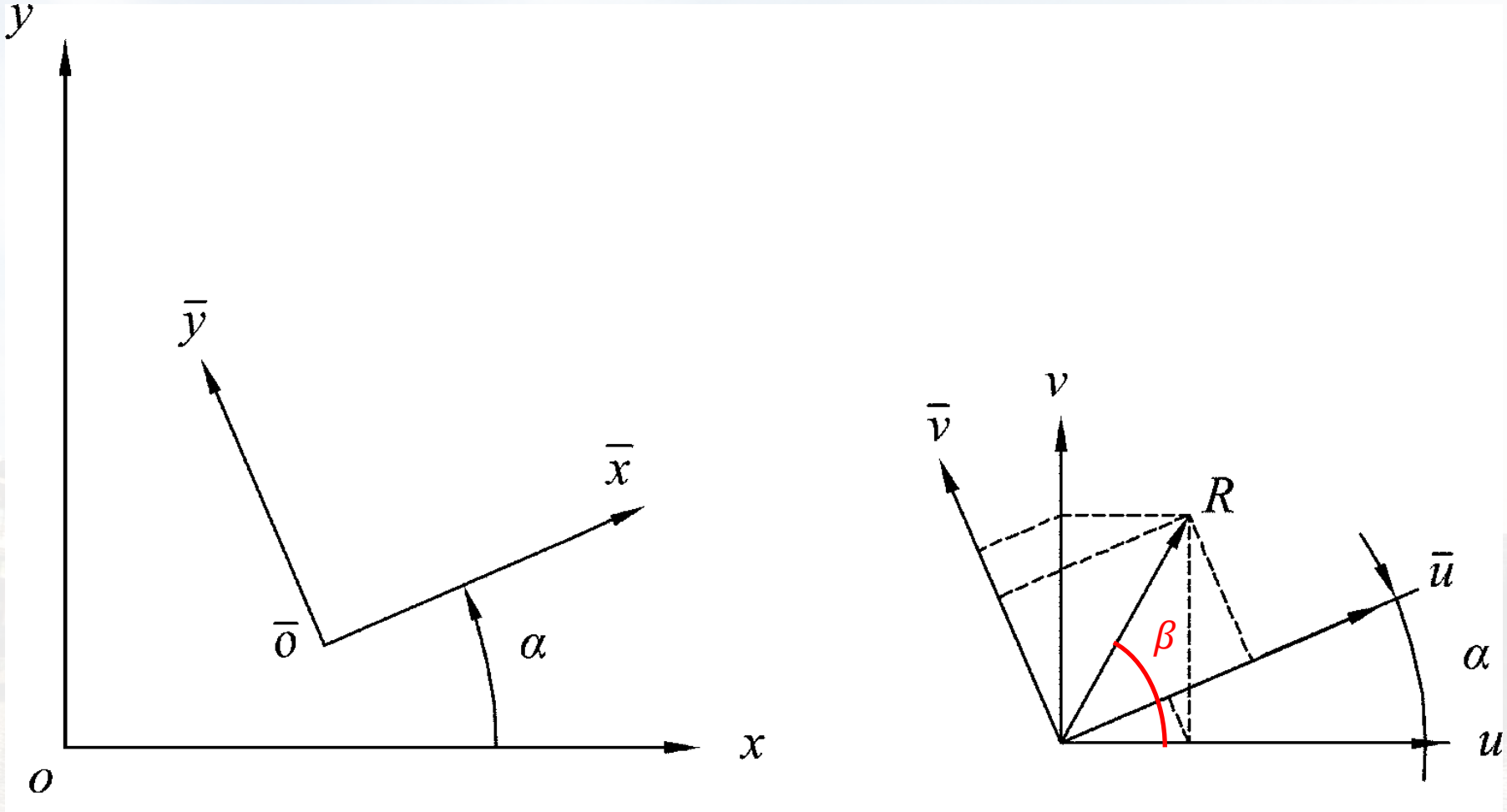
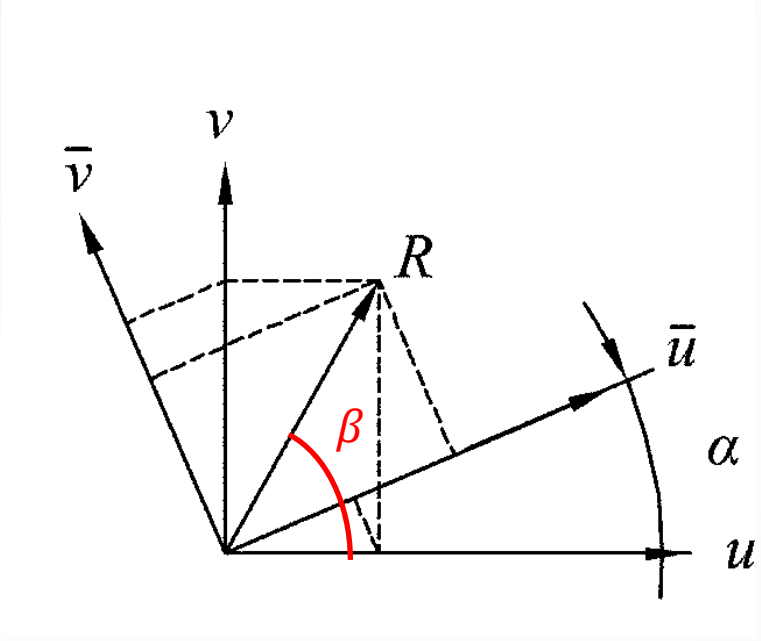


图2-4 局部坐标系与整体坐标系及其变换

$$\begin{cases} R\cos(\beta-\alpha)=\bar{u} \\ R\sin(\beta-\alpha)=\bar{v} \\ R\cos\beta=u \\ R\sin\beta=v \end{cases}$$

$$\begin{cases} \bar{u}=R(\cos\alpha\cos\beta+\sin\alpha\sin\beta)=u\cos\alpha+v\sin\alpha \\ \bar{v}=R(\sin\beta\cos\alpha-\cos\beta\sin\alpha)=v\cos\alpha-u\sin\alpha \end{cases}$$



$$\left.\begin{aligned} \bar{u}&=u\cos\alpha+v\sin\alpha \\ \bar{v}&=-u\sin\alpha+v\cos\alpha \end{aligned}\right\}$$

$$\begin{bmatrix} \bar{u} \\ \bar{v} \end{bmatrix}=\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} \qquad (2-9)$$

$$\begin{bmatrix} \bar{u} \\ \bar{v} \end{bmatrix} = \lambda \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\begin{bmatrix} \bar{u}_i \\ \bar{v}_i \\ \bar{u}_j \\ \bar{v}_j \end{bmatrix} = \begin{bmatrix} \lambda & | & 0 \\ - & + & - \\ 0 & | & \lambda \end{bmatrix} \begin{bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{bmatrix} \qquad (2-10)$$

$$\overline{\delta^e} = T\delta^e \qquad (2-11)$$

$$T = \begin{bmatrix} \lambda & | & 0 \\ - & + & - \\ 0 & | & \lambda \end{bmatrix} = \left[\begin{array}{cc|c} \cos\alpha & \sin\alpha & \\ -\sin\alpha & \cos\alpha & 0 \\ \hline \multicolumn{2}{c|}{0} & \begin{matrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{matrix} \end{array}\right] \qquad (2-12)$$

T为单位正交矩阵

$$\overline{F^e} = TF^e \qquad (2-13)$$

2.2.2.2 整体坐标系下的单元刚度矩阵

$$\overline{F}^{e}=\overline{K}^{e}\overline{\delta}^{e} \qquad (2-14)$$

$$TF^{e}=\overline{K^{e}}T\delta^{e} \qquad (2-15)$$

两边同时左乘T的转置，得到

$$T^{T}TF^{e}=T^{T}\overline{K^{e}}T\delta^{e} \qquad (2-16)$$

T为单位正交矩阵，所以T与T的转置乘积为单位矩阵，因此，

$$F^{e}=T^{T}\overline{K^{e}}T\delta^{e} \qquad (2-17)$$

$$K^{e}=T^{T}\overline{K^{e}}T$$

$$K^e=\begin{bmatrix} K_{ii}^e & K_{ij}^e \\ K_{ji}^e & K_{jj}^e \end{bmatrix}=\begin{bmatrix} \lambda^T & 0 \\ 0 & \lambda^T \end{bmatrix}\begin{bmatrix} \overline{K_{ii}^e} & \overline{K_{ij}^e} \\ \overline{K_{ji}^e} & \overline{K_{jj}^e} \end{bmatrix}\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$=\begin{bmatrix} \lambda^T\overline{K_{ii}^e}\lambda & \lambda^T\overline{K_{ij}^e}\lambda \\ \lambda^T\overline{K_{ji}^e}\lambda & \lambda^T\overline{K_{jj}^e}\lambda \end{bmatrix}$$

$$K_{ij}^e=\lambda^T\overline{K_{ij}^e}\lambda \qquad (2-18)$$

$$\begin{bmatrix} F_i^e \\ F_j^e \end{bmatrix} = \begin{bmatrix} K_{ii}^e & K_{ij}^e \\ K_{ji}^e & K_{jj}^e \end{bmatrix} \begin{bmatrix} \delta_i \\ \delta_j \end{bmatrix} \qquad (2-19)$$

$$\left.\begin{aligned} F_i^e &= K_{ii}^e \delta_i + K_{ij}^e \delta_j \\ F_j^e &= K_{ji}^e \delta_i + K_{jj}^e \delta_j \end{aligned}\right\} \qquad (2-20)$$

2.2.3 整体分析

1、等效节点荷载计算

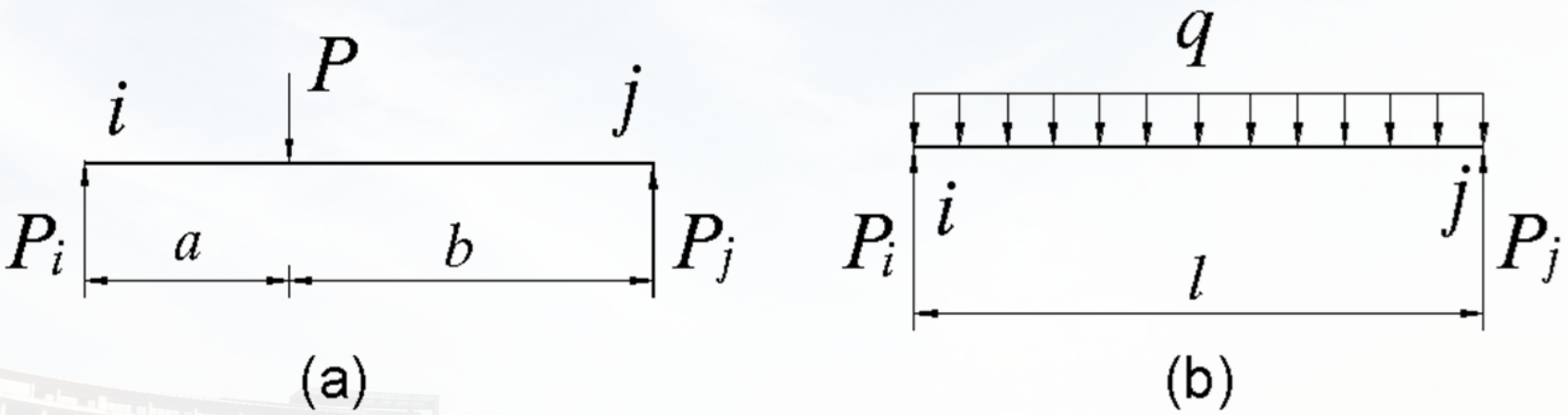


图2-5 集中力和均布载荷的等效节点载荷

$$P_i^e = \frac{b}{l}P \qquad P_j^e = \frac{a}{l}P$$

$$P_i^e = P_j^e = \frac{ql}{2}$$

2.2.3 整体分析

1、等效节点荷载计算

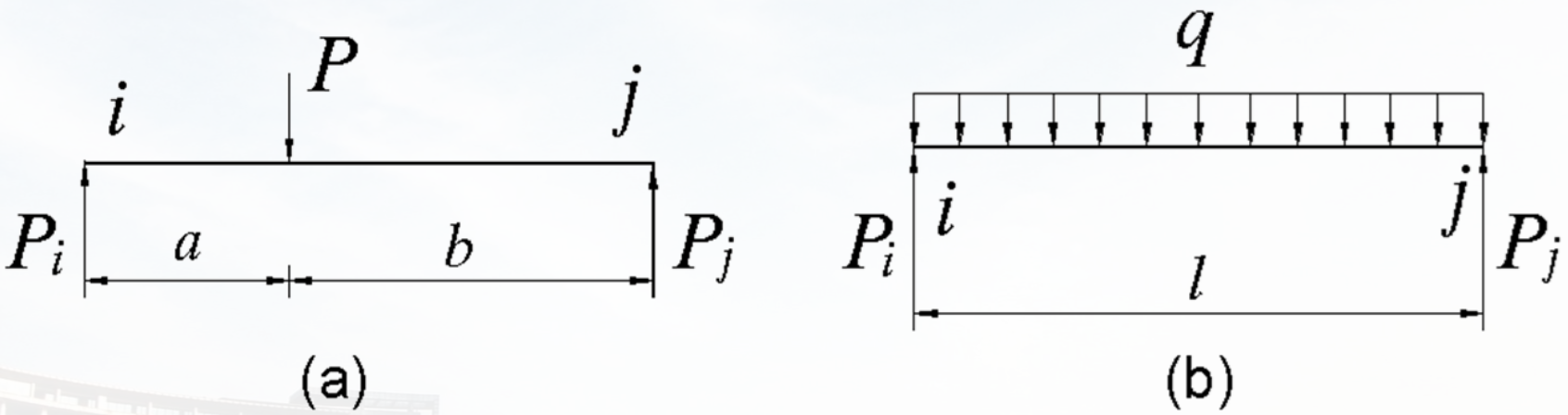


图2-5 集中力和均布载荷的等效节点载荷

$$P_i^e = \frac{b}{l}P \qquad P_j^e = \frac{a}{l}P$$

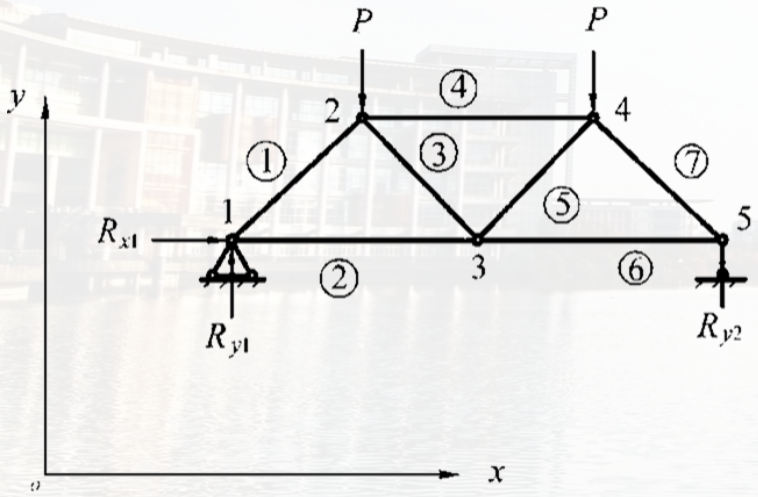
$$P_i^e = P_j^e = \frac{ql}{2}$$

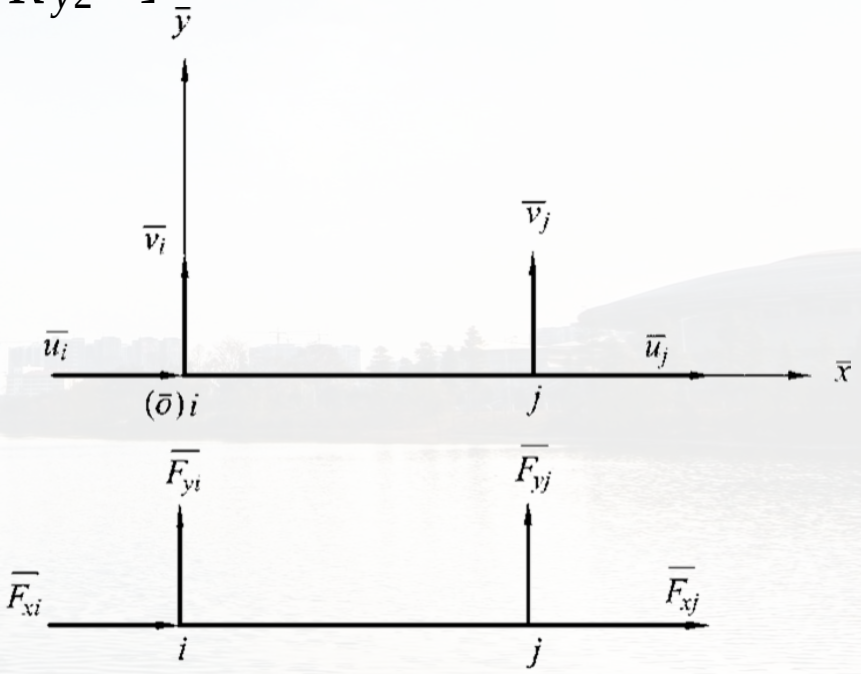
2、求整体刚度矩阵

$$\delta=[\delta_1\ \ \delta_2\ \ \delta_3\ \ \delta_4\ \ \delta_5]^T=[u_1\ \ v_1\ \ u_2\ \ v_2\ \ u_3\ \ v_3\ \ u_4\ \ v_4\ \ u_5\ \ v_5]^T$$

$$F=[P_1\ \ P_2\ \ P_3\ \ P_4\ \ P_5]^T=[P_{x1}\ \ P_{y1}\ \ P_{x2}\ \ P_{y2}\ \ P_{x3}\ \ P_{y3}\ \ P_{x4}\ \ P_{y4}\ \ P_{x5}\ \ P_{y5}]^T$$

$$=[R_{x1}\ \ R_{y1}\ \ 0\ \ -P\ \ 0\ \ 0\ \ 0\ \ -P\ \ 0\ \ R_{y2}\ ]^T$$





2.2 平面桁架

- 对于单元①

$$\begin{bmatrix} F_1^1 \\ F_2^1 \end{bmatrix} = \begin{bmatrix} K_{11}^{①} & K_{12}^{①} \\ K_{21}^{①} & K_{22}^{①} \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix}$$

- 对于单元②

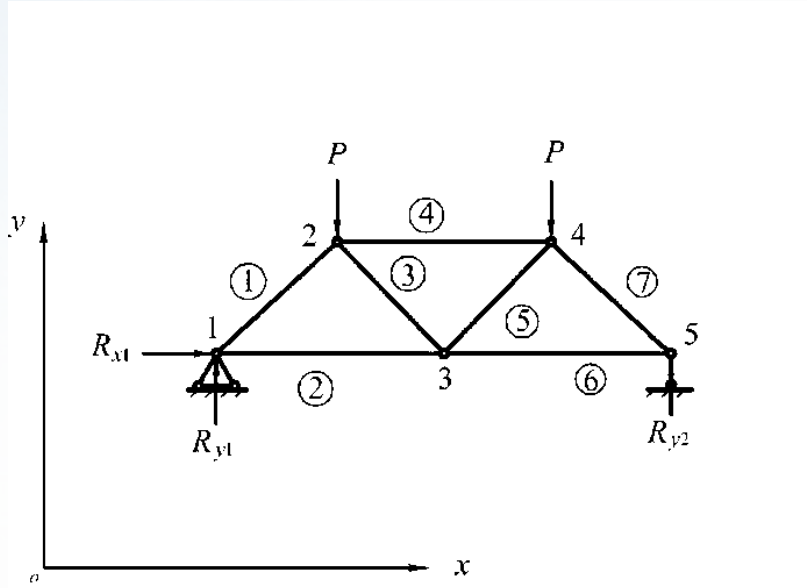
$$\begin{bmatrix} F_1^2 \\ F_3^2 \end{bmatrix} = \begin{bmatrix} K_{11}^{②} & K_{13}^{②} \\ K_{31}^{②} & K_{33}^{②} \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_3 \end{bmatrix}$$

- 对于单元③

$$\begin{bmatrix} F_3^3 \\ F_2^3 \end{bmatrix} = \begin{bmatrix} K_{33}^{③} & K_{32}^{③} \\ K_{23}^{③} & K_{22}^{③} \end{bmatrix} \begin{bmatrix} \delta_3 \\ \delta_2 \end{bmatrix}$$

- 对于单元④

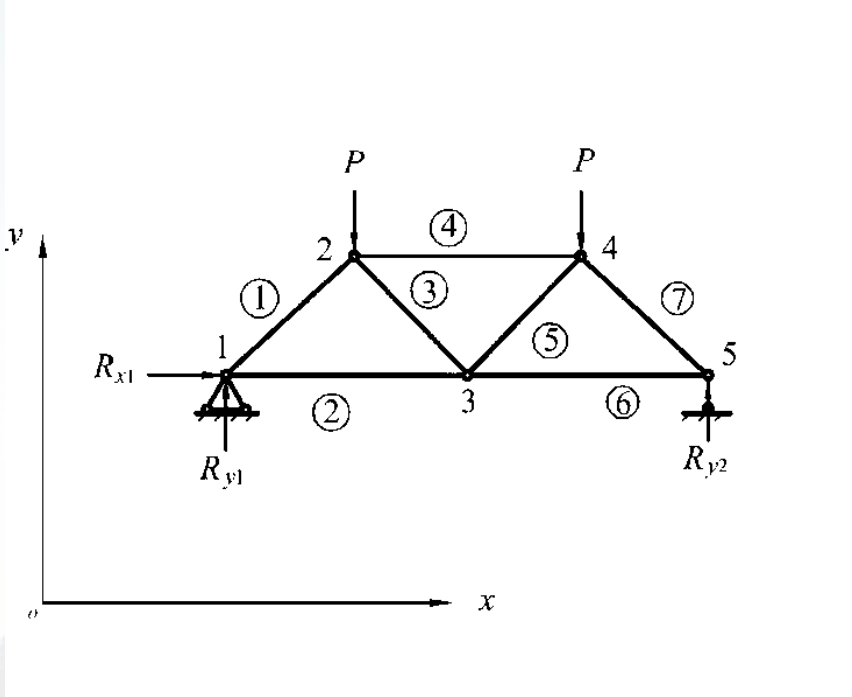
$$\begin{bmatrix} F_2^4 \\ F_4^4 \end{bmatrix} = \begin{bmatrix} K_{22}^{④} & K_{24}^{④} \\ K_{42}^{④} & K_{44}^{④} \end{bmatrix} \begin{bmatrix} \delta_2 \\ \delta_4 \end{bmatrix}$$



- 对于单元⑤ $$\begin{bmatrix} F_3^5 \\ F_4^5 \end{bmatrix} = \begin{bmatrix} K_{33}^{⑤} & K_{34}^{⑤} \\ K_{43}^{⑤} & K_{44}^{⑤} \end{bmatrix} \begin{bmatrix} \delta_3 \\ \delta_4 \end{bmatrix}$$

- 对于单元⑥ $$\begin{bmatrix} F_3^6 \\ F_5^6 \end{bmatrix} = \begin{bmatrix} K_{33}^{⑥} & K_{35}^{⑥} \\ K_{53}^{⑥} & K_{55}^{⑥} \end{bmatrix} \begin{bmatrix} \delta_3 \\ \delta_5 \end{bmatrix}$$

- 对于单元⑦ $$\begin{bmatrix} F_5^7 \\ F_4^7 \end{bmatrix} = \begin{bmatrix} K_{55}^{⑦} & K_{54}^{⑦} \\ K_{45}^{⑦} & K_{44}^{⑦} \end{bmatrix} \begin{bmatrix} \delta_5 \\ \delta_4 \end{bmatrix}$$



- 以节点2为例 $$\sum_3 -F_2^e + P_2 = 0 \qquad \sum_3 F_2^e = P_2$$

- 对于单元①:

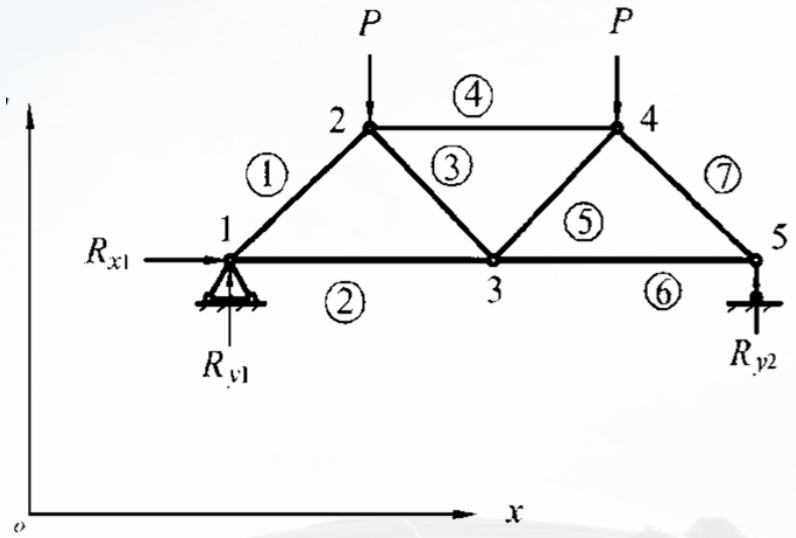
$$F_2^1 = K_{22}^{①}\delta_2 + K_{21}^{①}\delta_1$$

- 对于单元③:

$$F_2^3 = K_{22}^{③}\delta_2 + K_{23}^{③}\delta_3$$

- 对于单元④:

$$F_2^4 = K_{22}^{④}\delta_2 + K_{24}^{④}\delta_4$$



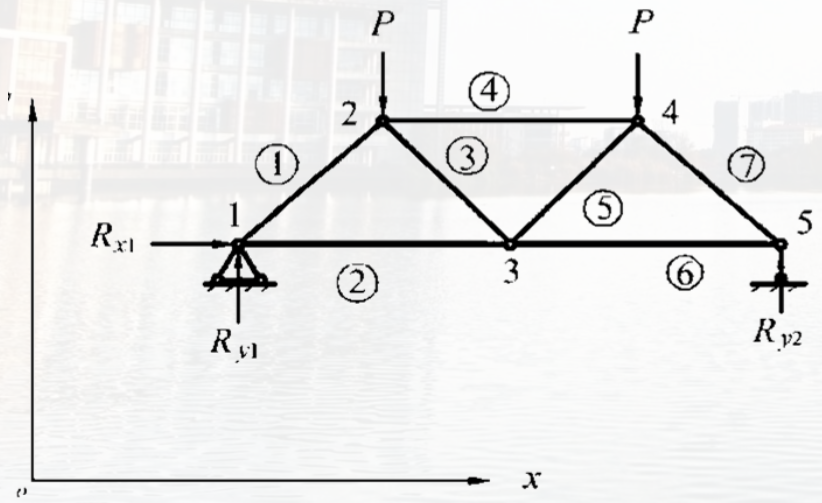
$$\left(K_{22}^{①} + K_{22}^{③} + K_{22}^{④}\right)\delta_2 + K_{21}^{①}\delta_1 + K_{23}^{③}\delta_3 + K_{24}^{④}\delta_4 = P_2$$

$$K=\begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} & K_{15} \\ K_{21} & K_{22} & K_{23} & K_{24} & K_{25} \\ K_{31} & K_{32} & K_{33} & K_{34} & K_{35} \\ K_{41} & K_{42} & K_{43} & K_{44} & K_{45} \\ K_{51} & K_{52} & K_{53} & K_{54} & K_{55} \end{bmatrix}=$$

$$\begin{bmatrix} K_{11}^{①}+K_{11}^{②} & K_{12}^{①} & K_{13}^{②} & 0 & 0 \\ K_{21}^{①} & K_{22}^{①}+K_{22}^{③}+K_{22}^{④} & K_{23}^{③} & K_{24}^{④} & 0 \\ K_{31}^{②} & K_{32}^{③} & K_{33}^{②}+K_{33}^{③}+K_{33}^{⑤}+K_{33}^{⑥} & K_{34}^{⑤} & K_{35}^{⑥} \\ 0 & K_{42}^{④} & K_{43}^{⑤} & K_{44}^{④}+K_{44}^{⑤}+K_{44}^{⑦} & K_{45}^{⑦} \\ 0 & 0 & K_{53}^{⑥} & K_{54}^{⑦} & K_{55}^{⑥}+K_{55}^{⑦} \end{bmatrix}$$

2.2 平面桁架

$$
\begin{bmatrix}
K_{11}^{①}+K_{11}^{②} & K_{12}^{①} & K_{13}^{②} & 0 & 0 \\
K_{21}^{①} & K_{22}^{①}+K_{22}^{③}+K_{22}^{④} & K_{23}^{③} & K_{24}^{④} & 0 \\
K_{31}^{②} & K_{32}^{③} & K_{33}^{②}+K_{33}^{③}+K_{33}^{⑤}+K_{33}^{⑥} & K_{34}^{⑤} & K_{35}^{⑥} \\
0 & K_{42}^{④} & K_{43}^{⑤} & K_{44}^{④}+K_{44}^{⑤}+K_{44}^{⑦} & K_{45}^{⑦} \\
0 & 0 & K_{53}^{⑥} & K_{54}^{⑦} & K_{55}^{⑥}+K_{55}^{⑦}
\end{bmatrix}
\times
\begin{bmatrix}
\delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \\ \delta_5
\end{bmatrix}
=
\begin{bmatrix}
P_1 \\ P_2 \\ P_3 \\ P_4 \\ P_5
\end{bmatrix}
$$



“刚度集成法”

整体刚度矩阵的性质

（1）是结构上的节点位移向量与节点荷载向量之间的转换矩阵。

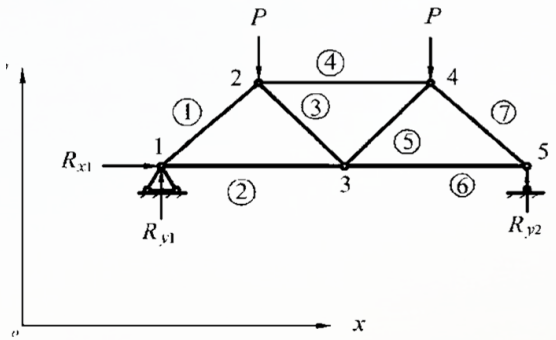
（2）矩阵中的元素均为刚度系数，其力学意义是：同一行所有元素是所有节点位移对同一个节点力的影响系数(或称贡献系数)；同一列的所有元素是同一个节点位移对所有节点力的影响系数。

（3）对称性

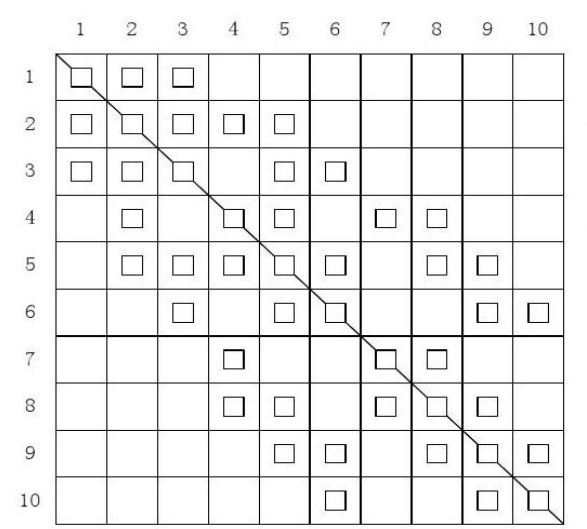
（4）分块性：处于主对角线上的子块称为主子块，其余子块称为副子块。主子块反映的是同一节点处的力与位移的关系，为正非零子块。副子块反映了结构上不同节点间的力与位移的相互关系。对于副子块，如果节点与节点在一个单元上，则是非零子块；如果节点与节点不在一个单元上，则就是零子块。

$$\begin{bmatrix} K_{11}^{①}+K_{11}^{②} & K_{12}^{①} & K_{13}^{②} & 0 & 0 \\ K_{21}^{①} & K_{22}^{①}+K_{22}^{③}+K_{22}^{④} & K_{23}^{③} & K_{24}^{④} & 0 \\ K_{31}^{②} & K_{32}^{③} & K_{33}^{②}+K_{33}^{③}+K_{33}^{⑤}+K_{33}^{⑥} & K_{34}^{⑤} & K_{35}^{⑥} \\ 0 & K_{42}^{④} & K_{43}^{⑤} & K_{44}^{④}+K_{44}^{⑤}+K_{44}^{⑦} & K_{45}^{⑦} \\ 0 & 0 & K_{53}^{⑥} & K_{54}^{⑦} & K_{55}^{⑥}+K_{55}^{⑦} \end{bmatrix} \times \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \\ \delta_5 \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ P_5 \end{bmatrix}$$

（5）稀疏性：即在中的非零元素集中在主对角线的周围，在其余位置上存在大量的零元素。非零的子块总数很少，如果节点个数越多，稀疏性越突出。利用刚度矩阵的稀疏性，可设法只存储非零元素，从而大大节省内存。



$$\begin{bmatrix} K_{11}^{①}+K_{11}^{②} & K_{12}^{①} & K_{13}^{②} & 0 & 0 \\ K_{21}^{①} & K_{22}^{①}+K_{22}^{③}+K_{22}^{④} & K_{23}^{③} & K_{24}^{④} & 0 \\ K_{31}^{②} & K_{32}^{③} & K_{33}^{②}+K_{33}^{③}+K_{33}^{⑤}+K_{33}^{⑥} & K_{34}^{⑤} & K_{35}^{⑥} \\ 0 & K_{42}^{④} & K_{43}^{⑤} & K_{44}^{④}+K_{44}^{⑤}+K_{44}^{⑦} & K_{45}^{⑦} \\ 0 & 0 & K_{53}^{⑥} & K_{54}^{⑦} & K_{55}^{⑥}+K_{55}^{⑦} \end{bmatrix} \times \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \\ \delta_5 \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ P_5 \end{bmatrix}$$



（6）奇异性：即刚度矩阵对应的行列式值为零，因此是奇异阵。

2.2.3.3 约束条件的引入

（1）划行划列法

$$\begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ P_5 \\ P_6 \end{bmatrix}$$

（2）主对角线元素置“1”法

$$\begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & 0 & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & 0 & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & 0 & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & 0 & k_{46} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ k_{61} & k_{62} & k_{63} & k_{64} & 0 & k_{66} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ 0 \\ P_6 \end{bmatrix}$$

（3）主元赋大值法

$$\begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & A & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ \delta_5^* A \\ P_6 \end{bmatrix}$$

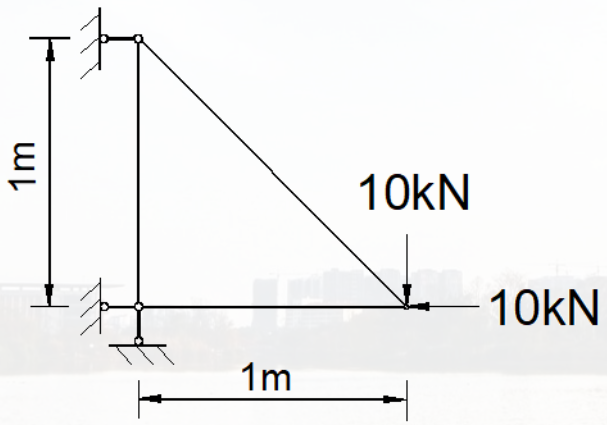
$$k_{51}X_1 + k_{52}X_2 + k_{53}X_3 + k_{54}X_4 + AX_5 + k_{56}X_6 = A\delta_5^*$$

$$X_5 = \delta_5^*$$

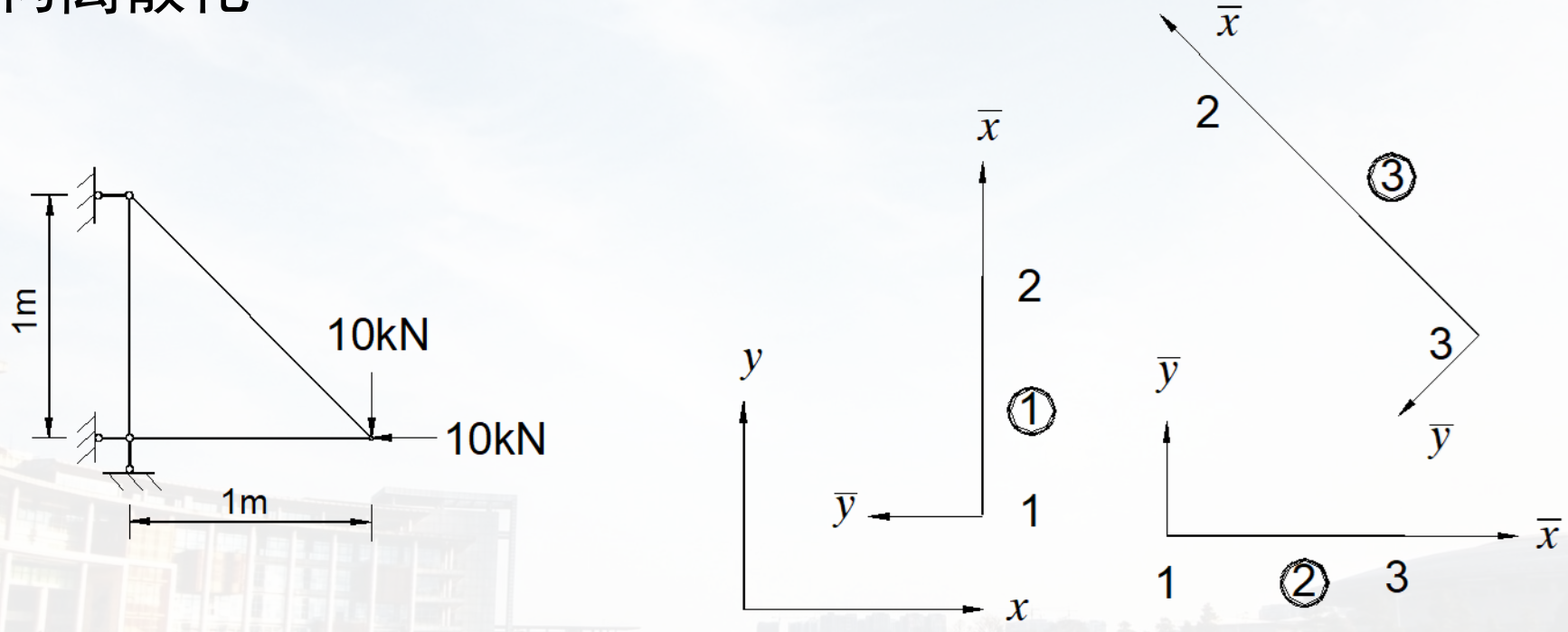
2.2.3.4 求单元内力和应力

$$\overline{\delta^e} = T\delta^e \qquad \overline{F^e} = \overline{K^e}\,\overline{\delta^e}$$

例题2-1 计算图2-6所示平面桁架的节点位移和杆件的内力。设各杆件的截面尺寸和制造材料均相同，其面积为 $A = 4\times10^{-2}\,m^2$，弹性模量 $E = 2.1\times10^{11}\,N/m^2$



1. 结构离散化



（a）平面桁架　　（b）单元、节点编号与整体、局部坐标系

图2-6 平面桁架及单元、节点编号与整体、局部坐标系

2. 计算单元在局部坐标系下的刚度矩阵

- 对于单元①和②

$$\frac{EA}{l}=8.40\times10^{9}\,N/m$$

$$\overline{K^{e}}=\frac{EA}{l}\begin{bmatrix}1&0&-1&0\\0&0&0&0\\-1&0&1&0\\0&0&0&0\end{bmatrix}$$

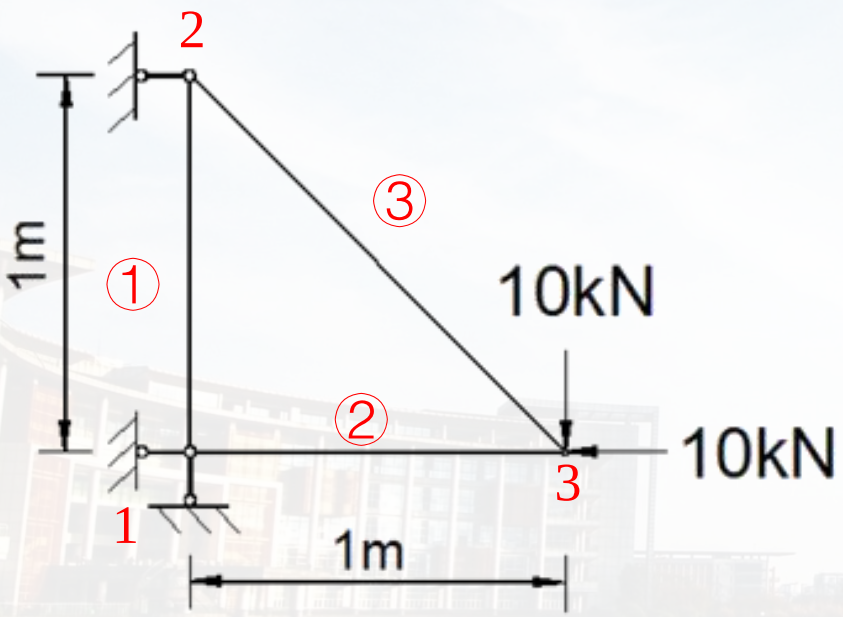
- 对于单元③

$$\frac{EA}{l}=5.9397\times10^{9}\,N/m$$

$$\overline{K^{3}}=5.9397\times10^{9}\times\begin{bmatrix}1&0&-1&0\\0&0&0&0\\-1&0&1&0\\0&0&0&0\end{bmatrix}$$

$$\overline{K^{1}}=\overline{K^{2}}=8.4\times10^{9}\times\begin{bmatrix}1&0&-1&0\\0&0&0&0\\-1&0&1&0\\0&0&0&0\end{bmatrix}$$

3. 计算单元在整体坐标系下的刚度矩阵



$$T=\begin{bmatrix}\lambda & | & 0\\ - & + & -\\ 0 & | & \lambda\end{bmatrix}=\left[\begin{array}{cc|cc}\cos\alpha & \sin\alpha & & \\ -\sin\alpha & \cos\alpha & & 0\\ \hline & 0 & \cos\alpha & \sin\alpha\\ & & -\sin\alpha & \cos\alpha\end{array}\right]$$

✓①单元与x正方向夹角90°

✓②单元与x正方向夹角0°

✓③单元与x正方向夹角135°

单元①

$$T^1=\begin{bmatrix} 0 & 1 & & \\ -1 & 0 & & O \\ & & 0 & 1 \\ & O & -1 & 0 \end{bmatrix}$$

$$K^1=8.4\times10^9\times\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$=8.4\times10^9\times\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

单元②

$$\boldsymbol{T}^2 = \boldsymbol{I}$$

$$K^{2} = 8.4\times10^{9}\times\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

单元③

$$\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$

$$T^3 = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & & & & \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & & & O & \\ & & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & & \\ & O & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & & \end{bmatrix}$$

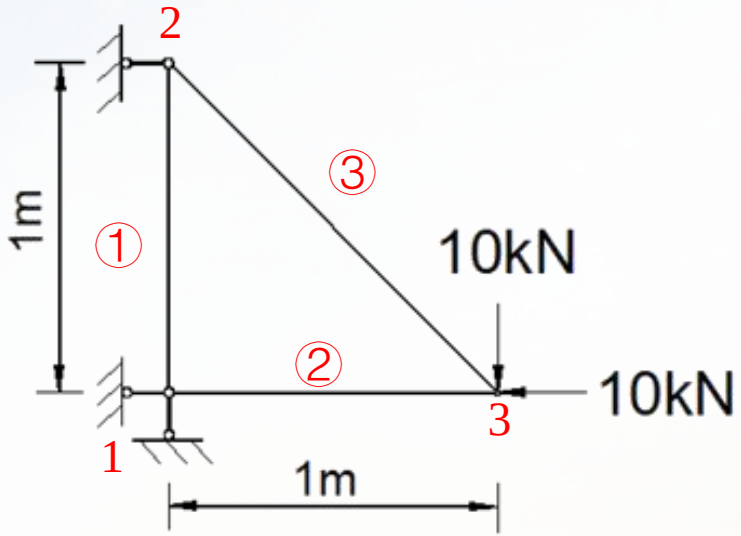
$$K^3 = 5.9397\times10^9\times\begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 0 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

$$= 2.9698\times10^9\times\begin{bmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

注意单元③的ij节点 $\begin{bmatrix} K_{33}^{③} & K_{32}^{③} \\ K_{23}^{③} & K_{22}^{③} \end{bmatrix}$

4. 有限元方程的建立

$$\begin{bmatrix} \boldsymbol{P}_1 \\ \boldsymbol{P}_2 \\ \boldsymbol{P}_3 \end{bmatrix} = \begin{bmatrix} \boldsymbol{K}_{11}^1 + \boldsymbol{K}_{11}^2 & \boldsymbol{K}_{12}^1 & \boldsymbol{K}_{13}^2 \\ \boldsymbol{K}_{21}^1 & \boldsymbol{K}_{22}^1 + \boldsymbol{K}_{22}^3 & \boldsymbol{K}_{23}^3 \\ \boldsymbol{K}_{31}^2 & \boldsymbol{K}_{32}^3 & \boldsymbol{K}_{33}^2 + \boldsymbol{K}_{33}^3 \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \end{bmatrix}$$



节点载荷向量

$$F = \left[F_{x1}, F_{y1} \mid F_{x2}, 0 \mid -10^4, -10^4 \right]^{\mathrm{T}}$$

节点位移向量

$$\delta = \left[0, 0 \mid 0, v_2 \mid u_3, v_3 \right]^{\mathrm{T}}$$

5. 引入边件条件

$$10^{6}\times\begin{bmatrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0\\0&0&0&11.3698&2.9698&-2.9698\\0&0&0&2.9698&11.3698&-2.9698\\0&0&0&-2.9698&-2.9698&2.9698\end{bmatrix}\begin{bmatrix}u_1\\v_1\\u_2\\v_2\\u_3\\v_3\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\\-10\\-10\end{bmatrix}$$

6. 求线性解方程组

$$\left.\begin{aligned}11.3698v_2+2.9698u_3-2.9698v_3&=0\\2.9698v_2+11.3698u_3-2.9698v_3&=-10\times10^{-6}\\-2.9698v_2-2.9698u_3+2.9698v_3&=-10\times10^{-6}\end{aligned}\right\}$$

$$v_2=-1.1905\times10^{-6}\qquad u_3=-2.3810\times10^{-6}\qquad v_3=-6.9388\times10^{-6}$$

7. 计算节点位移引起的节点载荷

$$\boxed{\overline{\delta^e} = T\delta^e} \qquad \boxed{\overline{F^e} = \overline{K^e}\,\overline{\delta^e}}$$

对于单元①

$$\begin{bmatrix} \overline{u_1} \\ \overline{v_1} \\ \overline{u_2} \\ \overline{v_2} \end{bmatrix} = T^1 \begin{bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1.1905 \end{bmatrix} \times 10^{-6} = \begin{bmatrix} 0 \\ 0 \\ -1.1905 \\ 0 \end{bmatrix} \times 10^{-6}$$

$$\begin{bmatrix} \overline{F^1_{x1}} \\ \overline{F^1_{y1}} \\ \overline{F^1_{x2}} \\ \overline{F^1_{y2}} \end{bmatrix} = \overline{K^1} \begin{bmatrix} \overline{u_1} \\ \overline{v_1} \\ \overline{u_2} \\ \overline{v_2} \end{bmatrix} = 8.4 \times 10^3 \times \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1.1905 \\ 0 \end{bmatrix} = \begin{bmatrix} 10.0 \\ 0 \\ -10.0 \\ 0 \end{bmatrix} \times 10^3$$

对于单元②

$$\begin{bmatrix} \overline{u_1} \\ \overline{v_1} \\ \overline{u_3} \\ \overline{v_3} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2.3810 \\ -6.9388 \end{bmatrix} \times 10^{-6}$$

$$\begin{bmatrix} \overline{F_{x1}^2} \\ \overline{F_{y1}^2} \\ \overline{F_{x3}^2} \\ \overline{F_{y3}^2} \end{bmatrix} = 8.4 \times 10^3 \times \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -2.3810 \\ -6.9388 \end{bmatrix} = \begin{bmatrix} 20.0 \\ 0 \\ -20.0 \\ 0 \end{bmatrix} \times 10^3$$

对于单元③

$$\begin{bmatrix} \overline{u_3} \\ \overline{v_3} \\ \overline{u_2} \\ \overline{v_2} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} -2.3810 \\ -6.9387 \\ 0 \\ -1.1905 \end{bmatrix} \times 10^{-6} = \begin{bmatrix} -3.2228 \\ 6.5900 \\ -0.8418 \\ 0.8418 \end{bmatrix} \times 10^{-6}$$

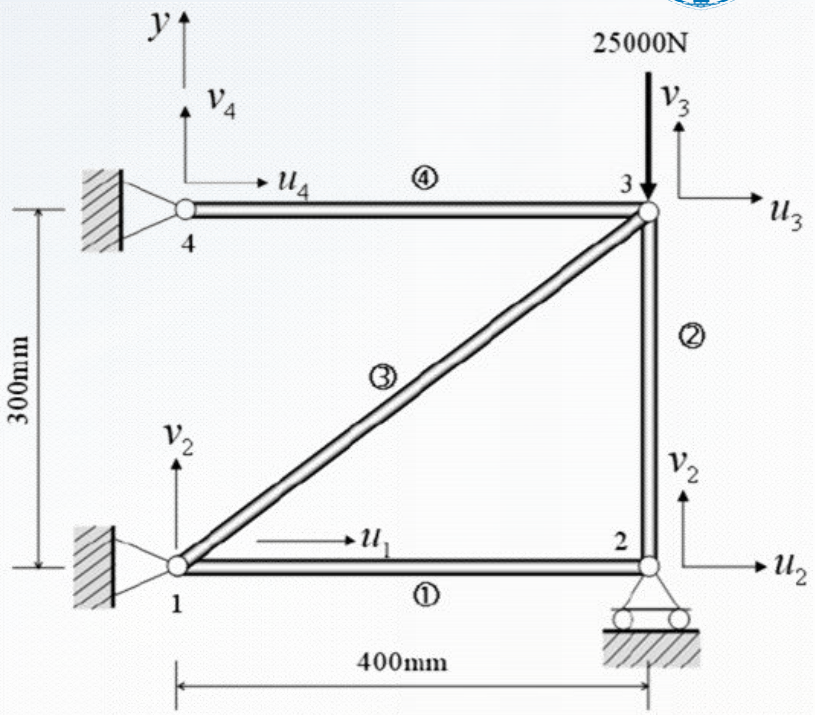
$$\begin{bmatrix} \overline{F_{x3}^{3}} \\ \overline{F_{y3}^{3}} \\ \overline{F_{x2}^{1}} \\ \overline{F_{y2}^{1}} \end{bmatrix} = 5.9397 \times 10^{9} \times \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -3.2228 \\ 6.5900 \\ -0.8418 \\ 0.8418 \end{bmatrix} \times 10^{-6} = \begin{bmatrix} -14.1424 \\ 0 \\ 14.1424 \\ 0 \end{bmatrix} \times 10^{3}$$

2.2 平面桁架

- Step1. 结构的离散化与编号

节点及坐标（对该结构进行自然离散）

节点	x	y
1	0	0
2	400	0
3	400	300
4	0	300



单元编号及对应节点

单元	节点1	节点2
①	1	2
②	3	2
③	1	3
④	4	3

各单元的长度及轴线方向余弦

单元	l	cos	sin
①	400	1	0
②	300	0	-1
③	500	0.8	0.6
④	400	1	0

2.2 平面桁架

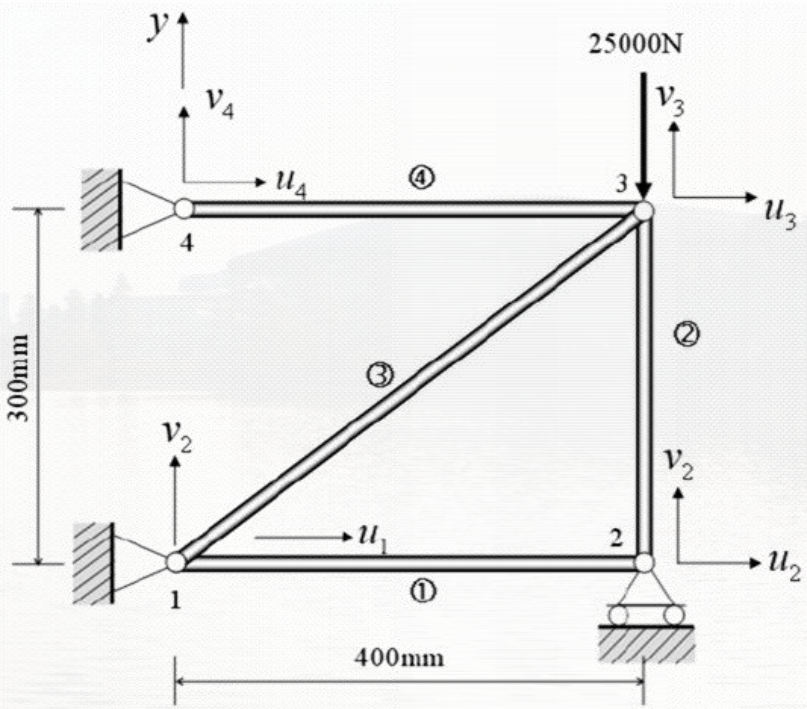
- Step2. 单元描述

$$K^{(1)}=\frac{29.5\times10^4\times100}{400}\begin{array}{c} \begin{array}{cccc} u_1 & v_1 & u_2 & v_2 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{array} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}\begin{array}{l} \leftarrow u_1 \\ \leftarrow v_1 \\ \leftarrow u_2 \\ \leftarrow u_2 \end{array}$$

$$K^{(2)}=\frac{29.5\times10^4\times100}{300}\begin{array}{c} \begin{array}{cccc} u_3 & v_3 & u_2 & v_2 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array}\begin{array}{l} \leftarrow u_3 \\ \leftarrow v_3 \\ \leftarrow u_2 \\ \leftarrow v_2 \end{array}$$

$$K^{(3)}=\frac{29.5\times10^4\times100}{500}\begin{array}{c} \begin{array}{cccc} u_1 & v_1 & u_3 & v_3 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{array} \\ \begin{bmatrix} 0.64 & 0.48 & -0.64 & -0.48 \\ 0.48 & 0.36 & -0.48 & -0.36 \\ -0.64 & -0.48 & 0.64 & 0.48 \\ -0.48 & -0.36 & 0.48 & 0.36 \end{bmatrix} \end{array}\begin{array}{l} \leftarrow u_1 \\ \leftarrow v_1 \\ \leftarrow u_3 \\ \leftarrow v_3 \end{array}$$

$$K^{(4)}=\frac{29.5\times10^4\times100}{400}\begin{array}{c} \begin{array}{cccc} u_4 & v_4 & u_3 & v_3 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{array} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}\begin{array}{l} \leftarrow u_4 \\ \leftarrow v_4 \\ \leftarrow u_3 \\ \leftarrow v_3 \end{array}$$



2.2 平面桁架

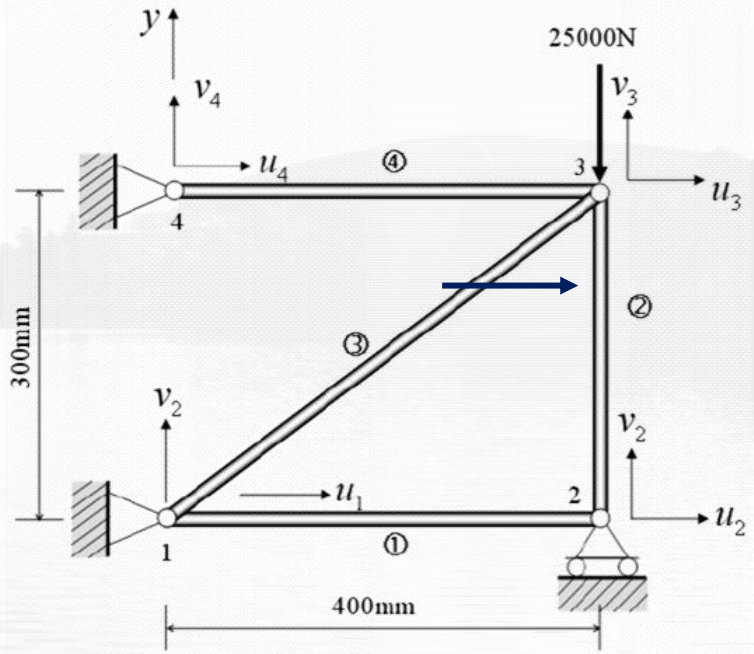
- Step2. 组装整体刚度方程
- 各个单元刚度矩阵/节点载荷按节点编号进行组装。

刚度矩阵：$\mathbf{K} = \mathbf{K}^{(1)} + \mathbf{K}^{(2)} + \mathbf{K}^{(3)} + \mathbf{K}^{(4)}$

节点位移：$\mathbf{q} = [u_1 \quad v_1 \quad u_2 \quad v_2 \quad u_3 \quad v_3 \quad u_4 \quad v_4]^T$

节点力：$\mathbf{P} = \mathbf{R} + \mathbf{F} = \begin{bmatrix} R_{x1} & R_{y1} & 2\times10^4 & R_{y2} & 0 & -2.5\times10^4 & R_{x4} & R_{y4} \end{bmatrix}^T$

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{bmatrix} = \begin{bmatrix} K_{11}^1 + K_{11}^3 & K_{12}^1 & K_{13}^3 & 0 \\ & K_{22}^1 + K_{22}^2 & K_{23}^2 & 0 \\ & & K_{33}^2 + K_{33}^3 + K_{33}^4 & K_{34}^4 \\ \text{对称} & & & K_{44}^4 \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}$$



整体刚度方程为：

$$
\frac{29.5\times10^4\times100}{6000}
\begin{array}{c}
\begin{array}{cccccccc}
u_1 & v_1 & u_2 & v_2 & u_3 & v_3 & u_4 & v_4 \\
\downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow
\end{array} \\
\begin{bmatrix}
22.68 & 5.76 & -15.0 & 0 & -7.68 & -5.76 & 0 & 0 \\
5.76 & 4.32 & 0 & 0 & -5.76 & -4.32 & 0 & 0 \\
-15.0 & 0 & 15.0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 20.0 & 0 & -20.0 & 0 & 0 \\
-7.68 & -5.76 & 0 & 0 & 22.68 & 5.76 & -15.0 & 0 \\
-5.76 & -4.32 & 0 & -20.0 & 5.76 & 24.32 & 0 & 0 \\
0 & 0 & 0 & 0 & -15.0 & 0 & 15.0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\end{array}
\begin{bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{bmatrix}
=
\begin{bmatrix} R_{x1} \\ R_{y1} \\ F_{x2} \\ R_{y2} \\ F_{x3} \\ F_{y3} \\ R_{x4} \\ R_{y4} \end{bmatrix}
=
\begin{bmatrix} R_{x1} \\ R_{y1} \\ 2\times10^4 \\ R_{yz} \\ 0 \\ -2.5\times10^4 \\ R_{x4} \\ R_{y4} \end{bmatrix}
$$

- Step2. 处理边界条件求解

- 边界条件BC(u): $u_1 = v_1 = v_2 = u_4 = v_4 = 0$

代入整体方程并化简得:

$$\frac{29.5 \times 10^4 \times 100}{6000}\begin{bmatrix} 15 & 0 & 0 \\ 0 & 22.68 & 5.76 \\ 0 & 5.76 & 24.32 \end{bmatrix}\begin{bmatrix} u_2 \\ u_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 2 \times 10^4 \\ 0 \\ -2.5 \times 10^4 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} u_2 \\ u_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0.2712 \\ 0.0565 \\ -0.2225 \end{bmatrix}$$

$$\Downarrow$$

所有节点位移: $\mathbf{q} = [0 \quad 0 \quad 0.2712 \quad 0 \quad 0.0565 \quad -0.2225 \quad 0 \quad 0]^T$

Thanks